Memoirs Of Memory

A look at the past (DDR) and insights into the present (DDR2): how RAM is shedding the shackles of bandwidth, and more

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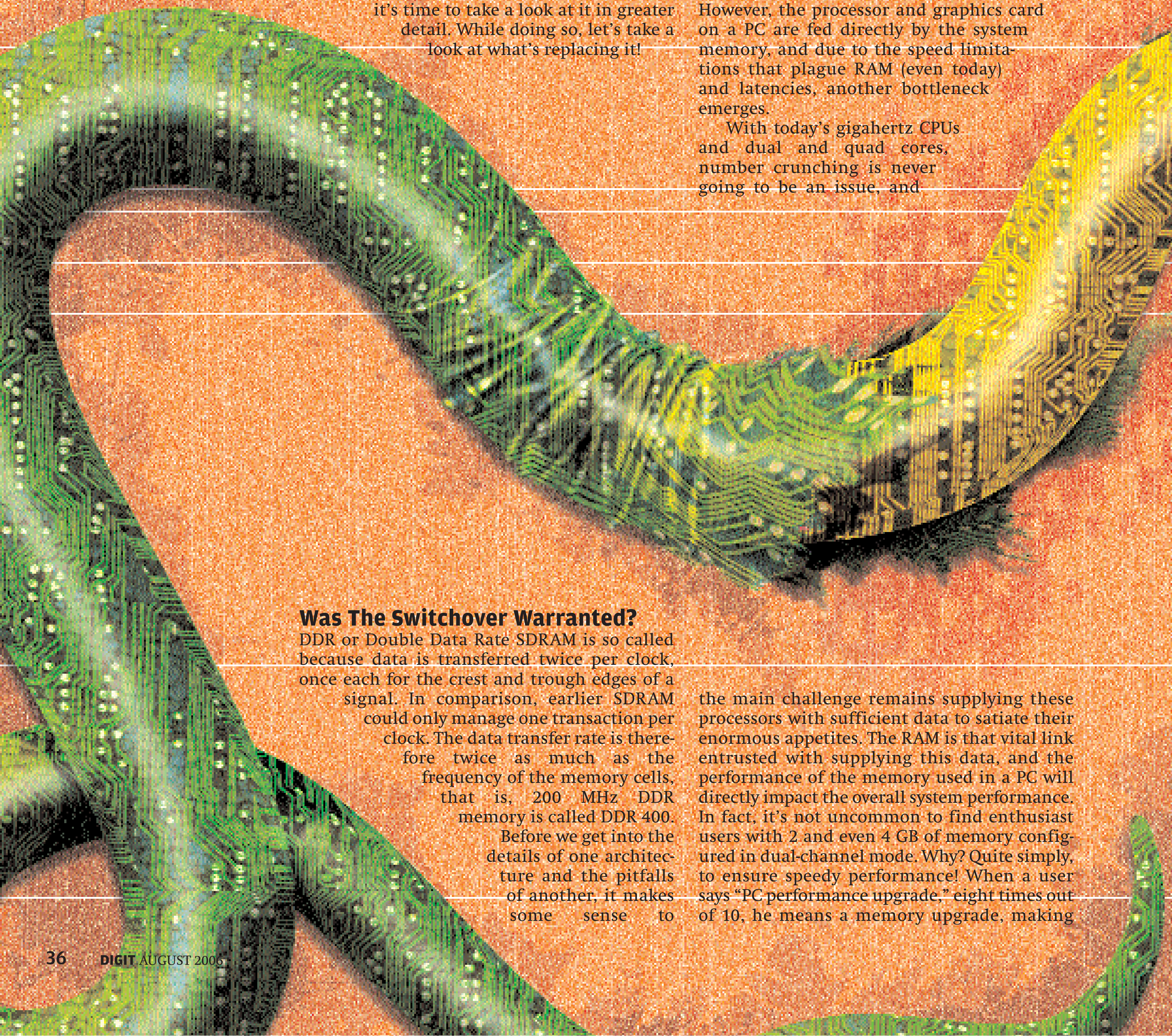
With AMD's jump on the DDR2 bandwagon, DDR SDRAM will ultimately find itself on the road to oblivion. Intel made the transition to DDR2 a few years ago with their 915/925 chipsets. With the end of the road for DDR memory clearly in sight, it's time to take a look at it in greater detail. While doing so, let's take a look at what's replacing it!



Was The Switchover Warranted?

DDR or Double Data Rate SDRAM is so called because data is transferred twice per clock, once each for the crest and trough edges of a signal. In comparison, earlier SDRAM could only manage one transaction per clock. The data transfer rate is therefore twice as much as the frequency of the memory cells, that is, 200 MHz DDR memory is called DDR 400.

Before we get into the details of one architecture and the pitfalls of another, it makes some sense to analyse why exactly the change was necessary. Let's look at the major problem all processors face today: the data transmission rate, or bandwidth. The performance of any PC depends upon just five major devices—the CPU, the chipset/motherboard, the RAM, the graphics solution and the hard disk(s). It's a known fact that the hard drives are by far the most serious bottleneck as far as data transmission goes. However, the processor and graphics card on a PC are fed directly by the system memory, and due to the speed limitations that plague RAM (even today) and latencies, another bottleneck emerges.

With today's gigahertz CPUs and dual and quad cores, number crunching is never going to be an issue, and the main challenge remains supplying these processors with sufficient data to satiate their enormous appetites. The RAM is that vital link entrusted with supplying this data, and the performance of the memory used in a PC will directly impact the overall system performance. In fact, it's not uncommon to find enthusiast users with 2 and even 4 GB of memory configured in dual-channel mode. Why? Quite simply, to ensure speedy performance! When a user says "PC performance upgrade," eight times out of 10, he means a memory upgrade, making